

2 (a) State what is meant by an ideal gas.

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..... [2]

(b) Ideal gases may be used to measure the temperature of the environment in a laboratory experiment. In some situations, however, it is preferable to use a thermocouple thermometer.

(i) Explain how a thermometer that uses an ideal gas can be advantageous compared with a thermometer that uses a liquid in glass.

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..... [1]

(ii) Describe a laboratory situation in which a thermocouple thermometer is preferable to a thermometer that uses an ideal gas.

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..... [1]

(c) A very small sample of an ideal gas consists of five molecules, each of mass 3.34×10^{-27} kg. At a particular instant, the speeds of the molecules are as shown in Figure 2.1.

357 ms ⁻¹	424 ms ⁻¹	509 ms ⁻¹	286 ms ⁻¹	435 ms ⁻¹
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Figure 2.1

Determine, to three significant figures, the root-mean-square (r.m.s.) speed c_0 of the molecules.

$c_0 =$ ms⁻¹ [2]

- (d) A larger sample of the gas in **2(c)** consists of 5.0×10^{23} molecules. These molecules have the same r.m.s. speed as the molecules of the gas in **2(c)**.

- (i) Calculate the amount of gas.

amount of gas = mol [2]

- (ii) Determine the thermodynamic temperature T_0 of the gas.

$T_0 =$ K [2]

- (iii) The temperature of the gas is now gradually increased.

On Figure 2.2, sketch the variation of the r.m.s. speed of the molecules of the gas with thermodynamic temperature.

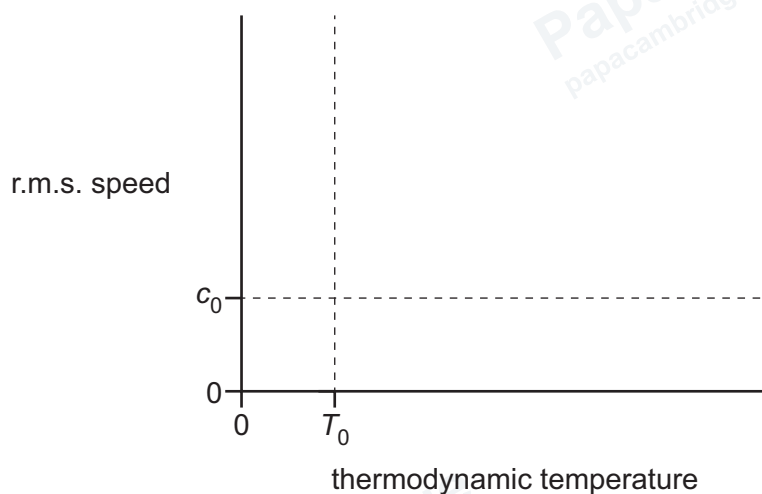


Figure 2.2

[3]

[Total: 13]